

Multi-Model Neural CDE Interpretability Audit Report

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Canonical conversion: From 3_model_comparison.md

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Models Compared: 3 Neural CDE Epochs 3,4 and 5

Samples Used: 3000

Random Seed: 42

Abstract

This report presents a multi-model interpretability, stability, and divergence audit of three Neural Controlled Differential Equation (Neural CDE) checkpoints “epochs” trained under identical architectural and data contracts. The analysis evaluates:

1. feature attribution consistency,
2. gradient sensitivity,
3. output stability,
4. information-theoretic divergence, and
5. surrogate explainability.

Physics-inspired metrics are used to characterize model coherence, response energy, and structural robustness across retraining epochs.

But, before I move forward, let me define what each one of those are, and their importance in measuring how one neural net compares against another.

Feature Attribution Consistency

Feature attribution consistency evaluates whether independently trained neural network models rely on the **same input features with similar relative importance** when producing predictions. This is typically assessed using techniques such as permutation importance or attribution mass comparisons across checkpoints. High consistency indicates that the learned representation of the input space is **structurally stable** and not sensitive to random initialization, optimizer noise, or stochastic training effects. In multi-model comparison, feature attribution consistency is critical because instability at this level signals spurious correlations, feature leakage, or overfitting, any of which can cause a model to perform well in backtests but fail catastrophically when deployed. Stable attribution implies that the model has learned **invariant, causal, or regime-robust signals** rather than transient statistical artifacts.

Gradient Sensitivity

Gradient sensitivity measures how responsive a model’s outputs are to **infinitesimal perturbations** in its inputs, typically via input–output gradients aggregated across samples,

features, or time. In continuous-time architectures such as Neural CDEs, gradients capture both **feature sensitivity and temporal sensitivity**, revealing how the model allocates attention across the input path. Comparing gradient sensitivity across models determines whether they react to market information in the **same directions and magnitudes**, even if their final outputs differ numerically. This evaluation is essential because two models can have similar predictions while exhibiting radically different internal sensitivities, implying hidden brittleness. High gradient alignment across models indicates **shared decision geometry**, while divergence may signal instability, over-parameterization, or sensitivity to noise.

Output Stability

Output stability evaluates the **statistical consistency of model predictions** across independently trained checkpoints when exposed to identical inputs. This includes measures such as correlation, mean squared deviation, variance ratios, distributional distance, and tail behavior. Output stability is particularly important in multi-head or multi-objective neural networks, where different outputs may be intentionally decorrelated but should remain **bounded and distributionally consistent** across retraining. In model comparison, output instability can indicate optimization path dependence, sharp minima, or latent representation drift. Stable outputs imply that the learned function approximator resides in a **robust region of function space**, increasing confidence that retraining or incremental updates will not materially alter system behavior.

Information-Theoretic Divergence

Information-theoretic divergence quantifies how different two models are in terms of the **distributions they induce**, rather than pointwise prediction errors. Metrics such as Jensen–Shannon divergence, Wasserstein distance, and rank correlation treat feature attributions or outputs as probability measures and compute the “distance” between them. This perspective is crucial in neural network comparison because it captures **global structural differences**, not just local discrepancies. Low divergence implies that retrained models encode essentially the same information content, while high divergence indicates representational drift or regime sensitivity. Information-theoretic analysis provides a mathematically principled way to assess whether models are **meaningfully different systems** or merely numerically distinct realizations of the same learned solution.

Surrogate Explainability

Surrogate explainability evaluates how well a complex neural network can be **approximated by a simpler, interpretable model**, such as a decision tree or linear rule set, over the same input–output mapping. Metrics like surrogate R^2 quantify how much of the neural model’s behavior can be captured by a low-complexity representation. In multi-model comparison, improving surrogate explainability across epochs indicates that the learned latent manifold is becoming **more structured, smoother, and more separable**, rather than more entangled. This is important because highly expressive models that cannot be approximated even locally are difficult to audit, validate, or govern. Strong surrogate explainability suggests that the neural network has converged toward **stable, human-compressible decision logic**, a key requirement for institutional deployment and long-term maintenance.

1. Executive Summary

- **Best Overall Model:** epoch_3_013026
- **Overall Health:**
 - Strengths Identified: 12
 - Concerns Flagged: 9

Key Cross-Model Findings

- Feature importance structure is **highly conserved** across all models.
 - Gradient saliency and temporal attention are **strongly aligned**.
 - Output heads exhibit **low correlation**, consistent with intentional multi-objective specialization.
 - Surrogate decision tree fidelity improves monotonically with epoch.
 - No evidence of catastrophic collapse or feature leakage is detected.
-

2. Configuration and Preprocessing Contract

2.1 Feature Registry

- **Feature Schema:** V2.2
- **Total Features:** 54
- **Input Dimension:** 54

2.2 Sequence Configuration

seq_len = 240

for all models:

- epoch_3_013026
- epoch_4_013026
- epoch_5_013026

2.3 Leakage Masking

The following features are explicitly zeroed to prevent forward-looking leakage:

target_spot = 0
max_dd_60m = 0

2.4 Scaling

- **Method:** Robust scaling using median and MAD
- **Clipping Range:**

$$[-10, +10]$$

- **NaN / Inf Handling:** Replaced with zero
-

3. Mathematical Foundations

3.1 Neural CDE Formulation

Let the latent state be:

$$Z_t \in \mathbb{R}^H$$

and the control path be:

$$X_t \in \mathbb{R}^D$$

The Neural CDE evolves according to:

$$dZ_t = f(Z_t; \theta) dX_t$$

where f is a learned vector field.

3.2 Integral Representation

$$Z_T = Z_0 + \int_0^T f(Z_t) dX_t$$

3.3 Discrete Approximation

$$Z_{t+1} = Z_t + f(Z_t) (X_{t+1} - X_t)$$

3.4 Vector Field Parameterization

$$f(z) = \tanh(W_2 \cdot \text{SiLU}(W_1 z + b_1) + b_2)$$

This bounds the vector field magnitude and enforces latent stability.

4. Permutation Importance

Feature importance is computed via repeated permutation:

$$I_j = \frac{1}{K} \sum_{k=1}^K [\mathcal{L}(f(X^{(j,k)}), y) - \mathcal{L}(f(X), y)]$$

Normalized importance:

$$\hat{I}_j = \frac{I_j}{\sum_{i=1}^D I_i} \times 100$$

5. Divergence Metrics

5.1 Cosine Similarity

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

5.2 Jensen–Shannon Divergence

$$D_{JS}(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M)$$

where:

$$M = \frac{1}{2}(P + Q)$$

5.3 Wasserstein Distance

$$W_1(P, Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)| dx$$

6. Gradient Saliency

Feature-wise saliency:

$$S_j = \frac{1}{N} \sum_{i=1}^N \left| \frac{\partial \hat{y}^{(i)}}{\partial x_j} \right|$$

Temporal saliency:

$$S_t = \sum_{j=1}^D \left| \frac{\partial \hat{y}}{\partial x_{t,j}} \right|$$

Gradient alignment between models A and B :

$$\rho_{\text{grad}} = \frac{\mathbf{g}_A \cdot \mathbf{g}_B}{\|\mathbf{g}_A\| \|\mathbf{g}_B\|}$$

7. Fisher Information

The Fisher Information Matrix is defined as:

$$\mathcal{F}_{ij} = \mathbb{E} \left[\frac{\partial \log p(y | x; \theta)}{\partial \theta_i} \frac{\partial \log p(y | x; \theta)}{\partial \theta_j} \right]$$

Diagonal empirical estimate:

$$\hat{\mathcal{F}}_{jj} = \frac{1}{N} \sum_{i=1}^N \left(\frac{\partial \mathcal{L}^{(i)}}{\partial \theta_j} \right)^2$$

8. Stability Metrics

8.1 Mean Squared Deviation

$$\text{MSD}(A, B) = \frac{1}{N} \sum_{i=1}^N \left(y_A^{(i)} - y_B^{(i)} \right)^2$$

8.2 Variance Ratio

$$\rho_{\text{var}} = \frac{\text{Var}(y_A)}{\text{Var}(y_B)}$$

8.3 Kolmogorov–Smirnov Statistic

$$D_{\text{KS}} = \sup_x | F_A(x) - F_B(x) |$$

9. Surrogate Decision Trees

Each model is approximated using a decision tree surrogate.

- **epoch_3_013026:**

$$R^2 = 0.6814$$

- **epoch_4_013026:**

$$R^2 = 0.7477$$

- **epoch_5_013026:**

$$R^2 = 0.8012$$

These values indicate increasing linear separability and interpretability across epochs.

10. Output Head Statistics (ROI Example)

For the ROI head:

$$\mu \approx -0.06, \sigma \approx 0.07-0.08$$

- Skewness: positive
- Kurtosis: sub-Gaussian

This reflects bounded downside with asymmetric tail behavior.

11. Pairwise Model Similarity

epoch_3 vs epoch_4

- Cosine Similarity: 0.983
- Spearman Correlation: 0.984
- Jensen–Shannon Divergence: 0.0716

epoch_3 vs epoch_5

- Cosine Similarity: 0.973
- Spearman Correlation: 0.977
- Jensen–Shannon Divergence: 0.0955

epoch_4 vs epoch_5

- Cosine Similarity: 0.998
 - Spearman Correlation: 0.996
 - Jensen–Shannon Divergence: 0.0368
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12. Interpretation Summary

- Feature learning is stable and convergent.
 - Differences between epochs are **magnitude-level**, not structural.
 - Output decorrelation is consistent with multi-objective design.
 - No evidence of feature leakage, collapse, or numerical instability.
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Figures & Exhibits Index — File Mapping

All plots and decision-tree diagrams are stored in:

spy-iron-condor-trading/reports/plots/

Exhibit Group F — Feature Attribution & Importance

Figure F-1 — Importance Comparison (All Models)

- **File:** importance_comparison.png
- **Status:** included
- **Used in:** Section 3, Section 4

Figure F-2 — Pairwise Feature Importance Matrix

- **File:** pairwise_matrix.png
- **Status:** included
- **Used in:** Section 4

Figure F-3 — Feature Importance Heatmap (epoch_3_013026)

- **File:** importance_heatmap_epoch_3_013026.png
- **Status:** included
- **Used in:** Section 3

Figure F-4 — Feature Importance Heatmap (epoch_4_013026)

- **File:** importance_heatmap_epoch_4_013026.png
- **Status:** included
- **Used in:** Section 3

Figure F-5 — Feature Importance Heatmap (epoch_5_013026)

- **File:** importance_heatmap_epoch_5_013026.png
- **Status:** included
- **Used in:** Section 3

Figure F-6 — Ranking Summary

- **File:** ranking_summary.png
- **Status:** included
- **Used in:** Section 4, Section 10

Exhibit Group G — Gradient & Sensitivity

Figure G-1 — Gradient Saliency (All Models)

- **File:** gradient_saliency.png
- **Status:** included
- **Used in:** Section 6

Figure G-2 — Gradient Alignment (Cross-Model)

- **File:** gradient_alignment.png
- **Status:** included
- **Used in:** Section 6

Exhibit Group S — Stability & Output Physics

Figure S-1 — Divergence Metrics Summary

- **File:** divergence_metrics.png
- **Status:** included
- **Used in:** Section 4

Figure S-2 — Stability (epoch_3 vs epoch_4)

- **File:** stability_epoch_3_013026_vs_epoch_4_013026.png
- **Status:** included
- **Used in:** Section 5

Figure S-3 — Stability (epoch_3 vs epoch_5)

- **File:** stability_epoch_3_013026_vs_epoch_5_013026.png
- **Status:** included
- **Used in:** Section 5

Figure S-4 — Stability (epoch_4 vs epoch_5)

- **File:** stability_epoch_4_013026_vs_epoch_5_013026.png
- **Status:** included
- **Used in:** Section 5

Figure S-5 — Output Distributions: call_offset

- **File:** output_dist_call_offset.png
- **Status:** included
- **Used in:** Section 5

Figure S-6 — Output Distributions: put_offset

- **File:** output_dist_put_offset.png
- **Status:** included

Figure S-7 — Output Distributions: wing_width

- **File:** output_dist_wing_width.png
- **Status:** included

Figure S-8 — Output Distributions: dte

- **File:** output_dist_dte.png
- **Status:** included

Figure S-9 — Output Distributions: pop

- **File:** output_dist_pop.png
- **Status:** included

Figure S-10 — Output Distributions: roi

- **File:** output_dist_roi.png
- **Status:** included

Figure S-11 — Output Distributions: max_loss

- **File:** output_dist_max_loss.png
- **Status:** included

Figure S-12 — Output Distributions: confidence

- **File:** output_dist_confidence.png
- **Status:** included

Figure S-13 — Output Distributions: entry

- **File:** output_dist_entry.png

- **Status:** included

Figure S-14 — Output Distributions: exit

- **File:** output_dist_exit.png
 - **Status:** included
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Exhibit Group T — Surrogate Explainability

Figure T-1 — Decision Tree (epoch_3_013026)

- **File:** decision_tree_epoch_3_013026.png
- **Status:** included
- **Used in:** Section 8

Figure T-2 — Decision Tree (epoch_4_013026)

- **File:** decision_tree_epoch_4_013026.png
- **Status:** included

Figure T-3 — Decision Tree (epoch_5_013026)

- **File:** decision_tree_epoch_5_013026.png
 - **Status:** included
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Exhibit Group I — Information Structure

Figure I-1 — Mutual Information

- **File:** mutual_information.png
- **Status:** included
- **Used in:** Sections 3, 6, 9

Figure I-2 — SHAP Comparison

- **File:** shap_comparison.png
 - **Status:** included
 - **Used in:** Sections 9, 10
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Exhibit Group H — Parameter Geometry (Fisher / Hessian)

Figure H-1 — Fisher Comparison (epoch_3 vs epoch_4)

- **File:** fisher_comparison_epoch_3_013026_vs_epoch_4_013026.png
- **Status:** included
- **Used in:** Section 7

Figure H-2 — Fisher Comparison (epoch_3 vs epoch_5)

- **File:** fisher_comparison_epoch_3_013026_vs_epoch_5_013026.png
- **Status:** included

Figure H-3 — Fisher Comparison (epoch_4 vs epoch_5)

- **File:** fisher_comparison_epoch_4_013026_vs_epoch_5_013026.png
- **Status:** included

Exhibit Group R — Model Summary

Figure R-1 — Model Radar

- **File:** model_radar.png
- **Status:** included
- **Used in:** Section 10

Summary

- **Total figures referenced:** 32
- **Total figures found in zip:** 32
- **Missing figures:** none
- **All exhibits fully accounted for and mapped 1-to-1**